Dataset README

Dataset — README

This file explains where the data included in this submission came from and how it was constructed. It accompanies the source code and the dissertation (see Chapter 5, “Implementation”, Section 5.3 for the full methodological account).

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1. Where the data was obtained

The historical market data underpinning this project was obtained from the **Colombo Stock Exchange (CSE)**, Sri Lanka’s national stock exchange.

The CSE does not provide a public, programmatic (API-based) interface for historical data. All data used in this project was therefore obtained through a **paid CSE premium data subscription**, which grants access to full historical Open, High, Low, Close and Volume (OHLCV) records for individual listed securities. Each security’s historical record was retrieved **individually** through the CSE subscription service, since no bulk or automated export was available.

This subscription authorises use of the data for **personal academic research**. In line with that licence:

- The raw subscription data is **not redistributed** in full anywhere in this submission or the deployed web application.
- Only the derived, cleaned dataset and the technical indicators computed from it are included here, for the purpose of reproducing this research.
- The deployed web interface exposes model outputs and a limited derived price series for chart display only — it does not expose the underlying subscription dataset.

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2. Companies included

Eighty (80) CSE-listed securities were selected for inclusion, chosen on the basis of data completeness (i.e. securities with a sufficiently long and continuous trading history to support the 30-day sequence construction described below). The full list of included securities is available in `data/company-list.json`.

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3. How the raw data was cleaned and consolidated

Each security’s file was retrieved separately from the CSE subscription portal, in the exchange’s own export format. The following steps were applied to consolidate and clean this data into a single master dataset:

1. **File discovery** — retrieved files were matched to their security symbol using a filename-pattern convention.
2. **Schema normalisation** — column names were standardised across all files into a common schema (Date, Open, High, Low, Close, Share_Volume).
3. **Date parsing** — the CSE’s exchange-specific textual date format was parsed into proper timestamp objects to guarantee correct chronological ordering across month and year boundaries.
4. **Quality correction**:
    - Missing opening prices were forward-filled from the preceding day’s closing price (reflecting days the security did not trade).
    - Missing low/high values were derived as the minimum/maximum of that day’s open and

close.
    - Zero share volume was retained as a genuine observation of market inactivity, not treated as missing data.
    - Records with no closing price were removed entirely, since no defensible value could be imputed for the field the target label depends on.
5. **Chronological sorting** — records were sorted by date within each security.

This process was applied consistently across all 80 securities and produced a consolidated master dataset of **323,998 daily records**.

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4. How the engineered dataset was built

From the cleaned OHLCV records, **30 technical indicators** were computed per security, spanning four categories:

- **Price-derived** — Daily/Log Return, High-Low Range, Open-Close Range
- **Trend** — Moving Averages (24/30/500-day), Parabolic SAR and SAR Trend
- **Momentum** — RSI, MACD, MACD Signal, MACD Histogram
- **Volatility** — Bollinger Bands (Upper/Lower/Middle/Width), 10-day realised volatility
- **Support/resistance** — Fibonacci retracement levels (6 levels)
- **Volume-derived** — Log Volume, Volume Moving Average, Volume Ratio

The binary directional target label was assigned per record: **1 (UP)** if the following trading day's closing price is strictly higher than the current day's close, **0 (DOWN)** otherwise. The final record of each security (which has no subsequent day to compare against) was excluded.

Full details of every indicator's formula, the exact cleaning rules, and the resulting class distribution (approximately 65.4% DOWN / 34.6% UP) are documented in Chapter 5 of the dissertation.

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5. How the chart image dataset was built

A separate visual dataset was generated **from the same cleaned master dataset** described above, by sliding a 30-day window across each security's price history (stride: 5 trading days) and rendering each window as an image in two encodings:

- **Candlestick charts** — a conventional candlestick rendering of the 30-day window, styled to match standard technical-analysis presentation.
- **Correlation heatmaps** — a Pearson correlation matrix of 14 selected features computed over the same 30-day window.

This produced a labelled corpus of **111,104 chart images** in total, carrying the identical directional label as the corresponding numerical sequence for that window. Full generation parameters (resolution, stride, colour conventions) are documented in Chapter 5, Section 5.4.

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6. What is and isn't included in this submission

Included: - `data/company-list.json` — the list of 80 CSE ticker symbols - `data/chart-data/*.json` — derived OHLC series per security, used to render historical charts in the deployed web interface - All source code (notebooks and web application) used to clean, engineer, train, evaluate and deploy the models

Not included (by design, per the subscription licence and file size): - The raw CSE subscription export files - The full 111,104-image chart corpus (regenerable from the master dataset using the code provided) - The full master dataset CSV with all 30 engineered features (available on request; regenerable from raw CSE data using the provided cleaning and featureengineering code, given a valid CSE subscription)

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7. Reproducing the dataset

Given access to raw CSE OHLCV exports for the same 80 securities, the complete master dataset and chart corpus can be regenerated by running the data preparation cells in the provided notebooks, in the order described in Chapter 5 of the dissertation and in the main project `README.md`.